

Rev B treats string tension as a **gate**, not a decoration. You pick a speaking length, a target pitch, and a real string's mass-per-length. The ideal-string formula then tells you the tension that string will pull. That number is compared against the planning ceiling (20 lbf per string / 100 lbf total) **and** against whatever the anchors, bars, spine, bolts, and tuners can actually carry. If any of those is unknown, the assembly stays untensioned.

Ideal string (what the document uses)

A flexible string fixed at both ends supports standing waves. The fundamental has wavelength $(2L)$, so

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

where

- f is fundamental frequency in hertz
- L is speaking length in metres (nut-to-bridge, not overall string length)
- T is tension in newtons
- μ is linear density in kg/m

Solve for the quantity you actually need:

$$T = \mu (2Lf)^2$$

or, if you already know a tension limit and want the string that would hit it,

$$\mu = \frac{T}{(2Lf)^2}$$

That is Mersenne's law. Wave speed on the string is $(v = \sqrt{T/\mu} = 2Lf)$. The document's source is Joe Wolfe's UNSW note **Strings, standing waves and harmonics**.

Same physics in the units string makers publish (D'Addario and similar charts):

$$T_{\text{lbf}} = \frac{UW}{(2L_{\text{in}}f)^2} \{386.4\}$$

UW is unit weight in lb/in. Multiply pounds-force by 4.448 to get newtons.

The Rev B worked example

The document does **not** select a string. It only shows what linear density would sit exactly on the planning ceiling.

Given:

- $(L = 32 \text{ in} = 0.8128 \text{ m})$
- $(f = 110 \text{ Hz})$ (A2)
- $(T = 20 \text{ lbf} = 88.9644 \text{ N})$

Wave speed:

$$v = 2Lf = 2 \times 0.8128 \times 110 = 178.816 \text{ m/s}$$

Then

$$\mu = \frac{T}{v^2} = \frac{88.9644}{178.816^2} = 2.7823 \times 10^{-3} \text{ kg/m} = 2.7823 \text{ g/m}$$

That is the number printed in the schematic update. A real string with that (μ) , at that length and pitch, would pull 20 lbf **if** it behaved as an ideal flexible string. Heavier string → more tension at the same pitch; lighter string → less.

Scaling you must not ignore

From $(T \propto \mu L^2 f^2)$:

| Change | Tension |

|---|---|

| Same string, same length, **double the pitch** | **4×** tension |

| Same string, same pitch, **double the length** | **4×** tension |

| Same pitch and length, **double (μ)** | **2×** tension |

| Octave down at same length | need **4×** (μ) (or a much heavier construction) |

Rev B states this explicitly because it is the usual way a “harmless” retune overloads the frame.

Why the formula is not enough to buy strings

The ideal law assumes:

- no bending stiffness
- uniform μ
- speaking length equal to the geometric nut-to-bridge distance
- small amplitude
- tension constant along the speaking length

Real strings violate all of those a little. Wound bass strings add mass without as much stiffness; plain wires are stiffer and slightly inharmonic; bearing edges and wrap-over at the nut/bridge shorten the *effective* L ; stretching while tuning changes μ slightly. That is why the document says use **manufacturer data at the exact pitch and scale**, then treat the ideal formula as a check, not a substitute.

Practical workflow for this build:

1. Measure the actual speaking length after the nut and bridge are in place (nominal 32 in, allowed 28–36 in; all five equal within 1/16 in unless you intend a scaled set).
2. Choose target pitches.
3. Look up each candidate string's unit weight or published tension at that scale and pitch, plus the maker's safe range.
4. Compute T (or read it from the chart). Convert to newtons for the load review.
5. Write the five rows of the string schedule: μ , L , f , T , and the limit source.
6. Sum the five operating tensions. Compare each string and the total to:
 - the planning ceiling (20 lbf / 100 lbf), **and**
 - the reviewed bar, bolt, spine, tuner, and joint capacities.

The 20 lbf / 100 lbf figures are **planning ceilings only**. They are not verified component ratings. Record whether a published number is a working-load limit or a breaking strength, and apply one documented design factor against the relevant failure mode.

How tension becomes a frame load

Rev B assumes the strings are approximately parallel and sit in one plane offset from the spine. At any section you check:

- axial force $\approx$ sum of the five tensions
- moment $\approx$ that resultant times the eccentricity e_i measured from **that section's reference axis**, not automatically from the spine face

Those loads go into bar bending, spine compression/bending, bolt bearing, joints, and base reactions. The vessel is not in this path. If a rating is unknown, you do not tension.

What belongs in the string schedule

Each row should be computable from measured (L) , chosen (f) , and a documented (μ) or chart tension:

$$T_i = \mu_i (2L_i f_i)^2$$

Then record $(\sum T_i)$ and the reviewed individual/component limits. Empty cells mean “unresolved,” not zero.

A quick numerical feel for the example pitch: 110 Hz at 32 in and 20 lbf wants $(\mu \approx 2.78 \text{ g/m})$. That is in the neighborhood of a fairly light low-tension string, not a standard guitar wound bass at concert tension. If you instead put a much heavier string at the same pitch, tension rises in proportion to (μ) and can blow past both the planning ceiling and the hardware.

If you want, I can next walk a filled string-schedule example (five specific products, pitches, and the resulting total load on the spine).

Here is a **filled planning example**, not a build release. It uses published D’Addario unit weights, the manufacturer formula, Rev B’s 32 in speaking length, and the 20 lbf / 100 lbf ceilings. Actual stock, measured (L) , and reviewed hardware limits still have to replace these numbers before anyone tensions the frame.

Formula used

$$T_{\{\text{lbf}\}} = \frac{\{\text{UW}\}}{(2L_{\{\text{in}\}}f)^2} \{386.4\}$$

$$T_{\{\text{N}\}} = 4.448 T_{\{\text{lbf}\}}, \quad \mu = \{\text{UW}\} \times \frac{0.45359237}{0.0254} \text{ kg/m}$$

For $(L=32)$ in this collapses to $(T_{\{\text{lbf}\}} = 10.6004 \frac{\{\text{UW}\}}{f^2})$.

Example configuration EX-B1

Assumptions written on the schedule so they can be challenged:

- Speaking length: **32.00 in (0.8128 m)** on all five strings
- Tuning: **A2–C3–E3–G3–C4** (A minor / C add, easy to hear intervals)

- A4 = 440 Hz equal temperament
- D'Addario UW from the official tension-chart booklet
- Planning ceilings only: 20 lbf/string, 100 lbf total
- No claim that tuners, bars, or the spine are rated for these loads

String	Note	f (Hz)	Product / construction	UW (lb/in)	μ (g/m)	T (lbf)	T (N)	Limit source
1	A2	110.00	D'Addario NW026 nickel roundwound .026	0.00012671	2.263	**16.25**	72.3	Chart UW + ideal law; 81% of 20 lbf ceiling
2	C3	130.81	D'Addario LE018 / PL018 plain steel .018	0.00007177	1.282	**13.02**	57.9	Same
3	E3	164.81	D'Addario LE014 / PL014 plain steel .014	0.00004342	0.775	**12.50**	55.6	Same
4	G3	195.00	D'Addario LE012 / PL012 plain steel .012	0.00003190	0.570	**12.99**	57.8	Same
5	C4	261.63	D'Addario LE010 / PL010 plain steel .010	0.00002215	0.396	**16.07**	71.5	Same

****Total operating tension: 70.8 lbf (315 N) — 71% of the 100 lbf planning ceiling.****

Final configuration ID: `EX-B1` Review reference: planning example only, not a connection review.

That is the table Rev B leaves blank. Each row is now calculable: product, μ , L , f , T , and the source of the limit.

Why these strings, not a guitar set

A guitar chart is almost always for ****25.5 in****. Same string, same pitch, longer scale:

$$\frac{T_{32}}{T_{25.5}} = \left(\frac{32}{25.5}\right)^2 \approx 1.57$$

So D'Addario's classical high E at 16.2 lbf on a guitar becomes ****25.5 lbf**** on this instrument. That single string already breaks the 20 lbf ceiling. You must recompute from UW at 32 in; you cannot copy the pounds printed on a guitar packet.

Strings that fail the same gate

Candidate at 32 in	Pitch	T	Why it is rejected
J3006 (classical low-E wound nylon)	A2	38.7 lbf	Almost 2× the per-string ceiling
J3005 (classical A wound nylon)	A2	23.6 lbf	Over 20 lbf

| NW030 nickel wound | A2 | 22.1 lbf | Over 20 lbf |
 | PL010 plain .010 | E4 | 25.5 lbf | Octave-up / guitar-E trap |

That is the document's "doubling pitch quadruples tension" rule in numbers: PL010 is fine at C4 (16.1 lbf) and illegal at E4 (25.5 lbf).

Nylon-family alternative (still only a plan)

If you want lower peak load and a softer broken-string hazard:

String	Note	Product	(T)
1	A2	J3004 SPC wound on nylon	13.6 lbf
2	C3	J2803 rectified nylon .040	8.4 lbf
3	E3	J2803 rectified nylon .040	13.3 lbf
4	G3	J2802 rectified nylon .032	11.8 lbf
5	C4	J2801 rectified nylon .028	15.0 lbf

Total **62.0 lbf (276 N)**. C3 is a bit slack at 8.4 lbf; that is a tone/stability choice, not a safety win by itself. Tuners still have to be rated for the chosen core (nylon vs steel).

What the 71 lbf does to the frame

Rev B's example geometry: spine depth toward the string plane ($d=3$) in, face-to-plane gap ($c=3$) in. For a rectangular 2×3 with the 3 in dimension toward the strings, the centroid sits 1.5 in behind the front face, so the string-plane eccentricity is about

$$e \approx 1.5 + 3 = 4.5 \text{ in}$$

For EX-B1, first-pass resultant and moment at mid-spine:

$$F_{\text{axial}} \approx 71 \text{ lbf}$$

$$M \approx 71 \times 4.5 = 320 \text{ lbf}\cdot\text{in} \approx 27 \text{ lbf}\cdot\text{ft}$$

Outer strings also sit ± 4 in sideways from the center string (2 in $\times$ 2 spaces). That sideways moment is smaller if the five tensions are similar, which they are here (12.5–16.3 lbf). The

vessel still carries **none** of this. Bar bending, bolt bearing, and base reactions are separate checks using the actual bar stock — which this example does not have.

How to use the schedule on the real build

1. Measure finished speaking length; do not keep 32.00 in if the nut and bridge landed elsewhere.
2. Keep or change the pitches; if you raise any pitch, recompute that row first.
3. Confirm the exact item number and UW from the current maker chart, not from this message.
4. Enter reviewed working-load limits for tuners, bars, bolts, and spine. If a cell is empty, that string stays slack.
5. Tension in the documented sequence, standing out of the string plane, with eye protection, and stop at 25 / 50 / 75 / 100% of the **reviewed** operating values — not automatically at 20 lbf.

EX-B1 only shows that a legal-looking five-string set exists under the planning ceiling. It does not release a cutting layout or certify the anchors.

The spine is a short column in **eccentric compression**. The five strings put an axial force (P) into it; because that force line sits in front of the wood axis, they also put a nearly constant moment $(M=Pe)$ into the length between the two anchor attachments. Bending, not pure crushing, is what sets the stress.

This uses example set EX-B1 $(P=70.8)$ lbf) and the Rev B planning ceiling $(P=100)$ lbf). It is a first-pass section check, not a connection review and not a release to cut or tension.

Model

Strings are taken as parallel. Each anchor bar then applies (P) and $(M=Pe)$ at the spine. Between those attachments the first-order moment is constant:

$$\sigma = \frac{P}{A} \pm \frac{Mc}{I} = \frac{P}{A} \pm \frac{Pe}{S}$$

- Front face (toward the strings): **compression** $(P/A + Pe/S)$
- Back face: $(P/A - Pe/S)$ (tension if $(e > S/A = d/2)$, which it is)

Two eccentricities, because Rev B says (e) is measured from the **section axis**, not automatically from the face:

Model	Formula	Meaning
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| String-plane (governing if the bars are stiff) | $(e = d/2 + c)$ | Resultant of the five strings. Rev B example uses $(c=3)$ in |
 | Face-attachment | $(e = d/2)$ | Force dumped at the front face only |

Attachment length is taken as $(L_e=32)$ in (speaking length). P-delta amplification is $(1/(1-P/P_{cr}))$ and is about **1.002** here, so it does not change the answer.

Section properties

Nominal (2×3) is (2.00×3.00) in. S4S dimensional stock is (1.50×2.50) in. Measure the actual blank.

[
 $A = bd, \quad I = \frac{bd^3}{12}, \quad S = \frac{bd^2}{6}$
]

Stock	Width (b)	Depth (d) toward strings	(A) (in ²)	(I) (in ⁴)	(S) (in ³)
Nominal (2×3)	2.00	3.00	6.00	4.50	3.00
Nominal (2×3)	3.00	2.00	6.00	2.00	2.00
Dressed (2×3)	1.50	2.50	3.75	1.95	1.56
Dressed (2×3)	2.50	1.50	3.75	0.70	0.94

Rev B's own clearance example used $(d=3)$ in, i.e. the larger dimension toward the strings. That is the better orientation: more (S) , and the 3 in depth is what the vessel-fit inequality already assumed.

Results at EX-B1 $(P=70.8)$ lbf

String-plane model, $(c=3)$ in:

Section	(e) (in)	$(M=Pe)$ (lbf·in)	(P/A) (psi)	(Mc/I) (psi)	Front (psi)	Back (psi)
Nom. (2×3) , $(d=3)$	4.50	319	12	106	118 C	95 T
Nom. (2×3) , $(d=2)$	4.00	283	12	142	154 C	130 T
Dressed, $(d=2.5)$	4.25	301	19	193	212 C	174 T
Dressed, $(d=1.5)$	3.75	266	19	283	302 C	265 T

If the bars only load the front face $(e=d/2)$, front stress drops to about **47 psi** on a full 2×3 and **76 psi** on dressed stock. Use that only if the connection calculation shows the string-plane couple is *not* delivered into the spine.

At the 100 lbf planning ceiling, scale every stress in the table by $(100/70.8 \approx 1.41)$. Worst dressed / weak-axis / string-plane case then reaches about **430 psi** compression on the front face.

What those numbers mean

Clear-wood averages at 12% MC (small, straight-grained lab specimens):

Species	MOR (bending ultimate)	Crushing parallel to grain	(E)
Hard maple	~15,800 psi	~7,830 psi	(1.83×10^6) psi
White ash	~15,000 psi	~7,410 psi	(1.74×10^6) psi

On the likely S4S piece with depth 2.5 in toward the strings, EX-B1 uses about **1.2%** of maple MOR in bending and **0.2%** of crushing strength in axial stress. Interaction:

$$\sqrt{\left(\frac{f_a}{F_{c,ult}}\right)^2 + \left(\frac{f_b}{\text{MOR}}\right)^2} \approx 0.015$$

So the **stick of wood**, as a prism, is not close to breaking at 71 lbf. That matches the low slenderness: Euler load about the bending axis is ~34,000 lbf for dressed $(d=2.5)$ in, ~79,000 lbf for a full 2×3. Operating load is 0.2% of that. Extra midspan bow from P-delta is ~0.01 in.

Graded-lumber allowables are much lower than clear-wood ultimates. NELMA red maple Select Structural is around $(F_b=1,300)$ psi and $(F_c=1,100)$ psi. Against those numbers the same dressed / string-plane case is about **15%** of (F_b) and **2%** of (F_c) . Still comfortable for the wood — if the blank is sound, straight, and actually that grade. An ungraded craft blank with a knot on the back (tension) face is a different member.

What this check does *not* cover

The spine stress is the easy part. The document's load review still has to do, with measured geometry:

- Orientation and actual section.** A dressed 2×3 with the 1.5 in dimension toward the strings doubles the bending stress relative to the Rev B $(d=3)$ sketch.
- How the bars attach.** Through-bolts, a head block, and any holes sit at the highest moment and cut the net section. Net-section bending and bolt bearing will exceed the mid-stick 200 psi long before the wood crushes.
- Weak-axis buckling.** The vessel does not brace the spine. For dressed stock the weak-axis Euler load is only ~12,000 lbf — still far above 71 lbf, but that assumes no crook and no side load from unequal outer strings.

4. **Creep and connections.** A 30-minute observation is not a long-term capacity. Hardwood screws in the tension path are already forbidden by Rev B.

5. **Sideways moment.** Outer strings sit ± 4 in off center. With EX-B1 tensions nearly even (12.5–16.3 lbf), that couple is small next to the 300 lbf-in forward moment. A one-sided heavy bass string would change that.

Practical takeaway

For EX-B1 on a sound hard-maple or white-ash 2×3, **mid-span bending stress** is on the order of 100–300 psi, front face in compression, back face in tension. The wood prism can carry that. Release still waits on the measured section, declared orientation, and the **bar-to-spine connection** calculation — that is where this frame will fail first, not in the middle of the stick.

The mid-span of the spine is the easy part. The **bar-to-spine joint** is where Rev B actually spends its caution, and the drawing is not a connection detail. What follows is a first-pass check of the two architectures the packet implies, using EX-B1 ($P=70.8$ lbf) and the 100 lbf planning ceiling. It is not a fabrication release.

What has to be connected

Upper and lower anchors each take the full string resultant and dump it into the spine.

EX-B1 string forces at $(x=-4,-2,0,+2,+4)$ in:

String	Note	(T) (lbf)	Arm from center
--- --- --- ---			
1	A2	16.25	-4 in
2	C3	13.02	-2 in
3	E3	12.50	0
4	G3	12.99	+2 in
5	C4	16.07	+4 in

Resultant ($P=70.8$ lbf), almost on center (plan moment only (0.8) lbf-in). The real eccentricity is **forward**: string plane $(c=2)-(3)$ in in front of the spine face, so $(e \approx d/2 + c \approx 4.25)$ in on dressed stock with the 2.5 in dimension toward the strings.

At each joint the spine must take:

- vertical force (P) (down at the top, up at the bottom) → **compression in the stick**
- couple $(M=Pe \approx 301)$ lbf-in → **front-face crushing + back-face reaction**, or **bolt prying**
- bar bending from the two outer strings, which sit (4) in off the spine centerline

Rev B forbids wood screws in this path. Through-bolts, washers, nyloc nuts, and a backing plate are the specified grammar.

Two architectures

The render and the text are not the same joint.

****A — Collar / yoke (what the picture shows).** ****** A metal ring clamps around the spine. The tuner bar or hitch bar is the front of that ring. The couple $\backslash(Pe\backslash)$ becomes a pair of horizontal forces on the front and back faces:

$$\backslash[\\H = \frac{P e}{d}\\backslash]$$

On dressed stock with $\backslash(d=2.5\backslash)$ in: $\backslash(H\approx 120\backslash)$ lbf. A 1.5 in tall collar on the 1.5 in front face gives about ****53 psi**** perpendicular-to-grain. Maple's proportional-limit crushing across the grain is ~1,470 psi; NDS-like allowables are ~620 psi. Face crushing is not the problem. ****Slip of the collar under 71 lbf of vertical load is.** ****** Friction alone is not a rated path. The collar still needs a shoulder, a through-pin, or through-bolts so the vertical force cannot walk.

****B — Front bar + through-bolts + back plate (what the parts list specifies).** ****** A bar carries the five holes. Two grade-marked bolts pass through bar, spine, and a steel backing plate. Vertical load is bolt shear and wood bearing parallel to grain. The couple $\backslash(Pe\backslash)$ tries to rotate the bar and ****pry the bolts in tension**** unless the bar also bears on a shoulder.

Those are the joints worth calculating. A bar that is only screwed to the front face is already excluded.

1. The bar itself

Rev B's correction exists because an 8 in bar cannot hold five holes at 2 in centers ****and**** positive end allowances.

$$\backslash[\\text{end-hole span}=(5-1)\times 2=8\backslash\text{in}\\backslash]$$

$$\backslash[\\L_{\text{bar}}=8+2e_{\text{end}}\\backslash]$$

End allowance $\backslash(e_{\text{end}}\backslash)$	Minimum bar length	Note
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0.50 in	9.0 in	2D for a ¼ in hole only
0.75 in	9.5 in	2D for a ⅜ in hole
1.00 in	10.0 in	Rev B: “geometry only, not approved”
1.50 in	11.0 in	Reasonable steel-plate edge

So the bar is **at least 11 in**, not 8 in. Length is the cheap part. Bending is not.

If the bar is held only across the spine width, the outer strings cantilever. For a 1.5 in wide dressed spine the left-side cantilever moment is **113 lbf·in** (the heavier A2 sits on that side). Treat that as the bar design moment until the bar is also supported outboard — which the vessel fit probably forbids.

Bending stress $\sigma = M/S$, $(S = wt^3/6)$ for a rectangular bar bent about its thickness:

Bar	S (in ³)	σ at 113 lbf·in	Fraction of yield
¼ × 1 in A36 flat	0.0104	10,800 psi	0.30 (F_y)
⅜ × 1 in A36	0.0234	4,820 psi	0.13 (F_y)
½ × 1 in A36	0.0417	2,710 psi	0.08 (F_y)
½ × 1½ in A36	0.0625	1,810 psi	0.05 (F_y)
½ × 1½ in brass	0.0625	1,810 psi	~0.10 of soft brass yield
¾ × 1½ in hard maple	0.141	800 psi	0.05 of clear MOR

A ¼ in plate works on paper at 30% of A36 yield **if** it is 1 in deep and has no holes in the high-moment zone. The five string holes sit exactly there. Net section and hole tear-out govern, not the gross S .

Practical bar for EX-B1: **½ × 1½ in A36 (or 304)**, 11–12 in long, holes bushed, end distance ≥ 2D. Brass of that size is acceptable at 71 lbf; do not use it if you later raise tension. A hardwood bar is possible but it then needs its *own* connection review and is a poor place to mount tuners.

2. Through-bolts

Assume two ⅜ in SAE Grade 5 bolts, nuts, washers, nyloc, backing plate. Maple ($G=0.63$).

Dowel bearing parallel to grain (NDS form):

$$F_{e\parallel} = 11200G \approx 7060 \text{ psi}$$

Raw bearing yield (F_{eDtm}) in 1.5 in maple with a ⅜ in bolt is about 4,000 lbf. The NDS Mode I reference value after the usual reduction is on the order of **1,000 lbf per bolt**.

Demand if the two bolts share (P) is **35 lbf** each. Steel single-shear ASD for a $\frac{3}{8}$ in Grade 5 shank is ~4,000 lbf. Shear and wood bearing under *pure* vertical load are not close.

The bolts do not see pure vertical load. If the bar is only a front plate, the couple $(Pe) \approx 301$ lbf·in is reacted by a vertical bolt gauge (h) :

$$T_{\text{pry}} = \frac{Pe}{h}$$

Vertical bolt spacing (h)	Tension per bolt from the couple
1.0 in	301 lbf
1.5 in	201 lbf
2.0 in	150 lbf
2.5 in	120 lbf

Each bolt then has **tension + shear**. A $\frac{3}{8}$ in Grade 5 still has tensile area to spare (~0.077 in², proof load thousands of pounds), but:

- A dressed 2×3 is only 2.5 in on the long face. Two bolts stacked 2 in apart with washers and edge distance barely fit.
- Through a 1.5 in face, a $\frac{3}{8}$ in bolt has only 0.75 in to the edge if centered. NDS loaded-edge distances of $1.5D-4D$ are **0.56–1.50 in**. A $\frac{3}{8}$ in bolt is the largest that is even arguable in 1.5 in thickness; $\frac{1}{2}$ in is too big.
- Two bolts side-by-side in the 1.5 in width: $(2 \times 0.375 = 0.75)$ in of hole in 1.5 in of wood. Do not do that.

So the viable pattern on S4S 2×3 is **two $\frac{5}{16}$ or $\frac{3}{8}$ in bolts, stacked up the spine (Z), not across it**, plus a full-face steel back plate — not a lone washer. A 1 in USS washer on the back face is 112 psi under 71 lbf, which is legal but a sloppy way to carry a 300 lbf·in couple.

3. Spine net section at the joint

This is the first place the stick of wood actually feels the joint.

Two $\frac{3}{8}$ in holes stacked vertically, dressed 1.5 × 2.5 in, holes through the 2.5 in depth:

- One hole in a given cross-section $\rightarrow (A_{\text{net}} \approx 2.81)$ in²
- Axial stress from 71 lbf $\approx$ **25 psi**
- Bending from (Pe) still uses nearly the full (S) , minus a local hole. Stress raiser, not a capacity problem at 71 lbf

The joint does not fail the spine in net-section tension/compression at these loads. It can split the spine if edge distance is short or if the collar crushes an arris. That is a detailing issue, not a 15,800 psi MOR issue.

4. Upper vs lower joints are not identical

****Upper.**** Tuners add a second load path. If the machines are on a separate head block, the **block-to-spine** joint is the tension connection; the bar may only locate the nut. If the tuners sit in the bar, the bar must take winding torque, broken-string shock, and the 113 lbf-in bend. Enclosed bass or zither pins need a declared mounting thickness — often thicker than a ½ in steel flat. A hardwood head block through-bolted to the spine, with the metal bar as a wearing strip, matches both the picture and the parts list.

****Lower.**** Ball-end pins or hitch pins in the bar, bar through-bolted to the spine or captured in the base shoe. The base already takes compression from the spine's end grain. Do not ask the same small bolts to also be the anti-tip restraint; Rev B separates those load cases.

****Both ends.**** Guards over cut string ends. Stand out of the string plane while tensioning. Inspect at 25 / 50 / 75 / 100% of the reviewed operating tension, not automatically at 20 lbf.

What fails first if you get this wrong

At EX-B1 loads the ranking is:

1. ****Undersized bar length**** — hole blows out the end (the Rev A defect).
2. ****Thin bar + five holes**** — net-section yield or fatigue at the outer holes, especially if someone later tunes up a fifth.
3. ****Collar with no positive vertical stop**** — joint slips, clearance to the vessel disappears, glass gets loaded. That is the dangerous one.
4. ****Bolts too large for a 1.5 in face**** — split the spine.
5. ****Washer instead of a back plate**** — local perp-grain crush and rotation under the couple.
6. Mid-stick bending of the spine (~200 psi) — last.

None of items 1–5 are certified by the 200 psi spine-stress calculation.

A connection that is consistent with Rev B

Call it planning detail ****CX-1****, still not a cutting drawing:

- Bar: ½ × 1½ × 12 in A36 or 304, five holes at 2.00 in centers, end distance 2.0 in, holes bushed.
- Spine: hard maple or white ash, ****actual**** section recorded, ****larger dimension toward the strings****.

- Two $\frac{3}{8}$ -16 Grade 5 through-bolts, stacked vertically ≥ 1.5 in apart, holes drilled for shank fit, not sloppy.
- $\frac{1}{8}$ in steel back plate covering the back face at the joint, nyloc nuts, flat washers under head and nut, torque marked at T0.
- Collar or shoulder so the 71 lbf vertical load cannot rely on friction.
- Separate head block for tuners if the tuner footprint does not fit the $\frac{1}{2}$ in bar.
- Same arrangement, inverted, at the lower anchor.
- No wood screws in this path.

On EX-B1 that joint is lightly stressed. It is still the member you design, because every millimetre of slip here is a millimetre of vessel interference, and because a broken-string jerk loads the same bolts that look idle at 35 lbf of shear.

Measure the real spine, pick the real bar and bolt lot, and fill the "Bar material and calculated connection limits" row of the worksheet. Until that row has numbers and a review date, Rev B says the strings stay slack.